

FS13.OC.12

Fine-scale spatial patterns and pond morphology jointly affect metacommunity structure of Neotropical tadpoles

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There is a contentious debate about the scale at which spatial and environmental processes influence the variation in species composition (beta diversity) among sites. Much of this debate is related to the discussion about the role of neutral and niche processes in community ecology. Accordingly, there is no consensus if dispersal-limited organisms are strongly influenced by spatial dynamics at fine spatial scales. We analyzed the relative influence of pond morphology, water chemistry, and spatial variables on a tadpole metacommunity from southeastern Brazil. We asked the following questions: 1) What is the relative influence of water chemistry and pond morphology to the variation in species composition of tadpoles across ponds?; 2) Are fine-scale spatial processes influencing metacommunity structure of tadpoles? Since amphibians are noticeably dispersal-limited, we predict that spatial variables or the joint effect of space and environment will have a larger influence on species composition of tadpoles than environmental variables alone (water chemistry and pond morphology). We sampled tadpoles and environmental variables between August 2008 and January 2009 in 13 water bodies in the Serra da Bocaina National Park. We partitioned the variation in species composition among water chemistry, pond morphology, and spatial variables (PCNM axes) using partial RDA analysis in the R software. We found that spatial processes are important even at fine scale ($\sim 11 \text{ Km}^2$) and that pond morphology strongly affected metacommunity structure. Apparently, adult anurans reproductive behavior is driving metacommunity dynamics, since pond morphology have been shown to affect breeding site selection by adults, but not water chemistry. If local amphibian communities are restricted to habitat patches and have low dispersal abilities, the limited gene flow among communities may favor local extinction more often than previously thought. Our results present important implications for understanding amphibian species turnover in tropical environments.